Notes on the Regeneration of the Fore-Limb in various Genera of Mantidae.

By

A. Landsborough Thomson, O.B.E., M.A., D.Sc.

(Aus der Biologischen Versuchsanstalt der Akademie der Wissenschaften in Wien, Zoologische Abteilung.)[1])

(Eingegangen am 2. Juli 1921.)

List of Contents.

General.

The following notes relate to experiments carried out during the earlier part of the summer of 1914 at the Biological Research Institute in Vienna. The subject of these experiments was suggested by Professor Przibram, and they indeed form a slight supplement to well-known work of his own. For the kind supervision exercised by Professor Przibram throughout the course of the experiments, and for the supply of material and the opportunities for work afforded by the Institute, I wish gratefully to acknowledge my indebtedness. Unfortunately the events of August, 1914, compelled me to bring my work to a hurried conclusion and were also the cause of the loss of many of my preserved specimens; several points, therefore, remain much less completely worked out than one could have wished.

---

[1]) Ein Auszug dieser Arbeit mit gleichlautendem Titel erschien als Mitteilung Nr. 55 aus der Biologischen Versuchsanstalt der Akademie der Wissenschaften, Zoologische Abteilung, Vorstand H. Przibram, im Akademischen Sitzungsanzeiger Wien, Nr. 7 u. 8, 1921.

I am also indebted to my father, Professor J. Arthur Thomson, L.L.D., University of Aberdeen, and to my friend Dr. George Riddoch, for helpful suggestions.

Statement of Problem.

The Mantidae are characterised by their specialised predatory fore-limbs as contrasted with the walking legs which form the middle and hind pairs. The fore-limbs are very strong and are not subject to loss by autotomy, whereas the other legs are more slender and readily autotomise between the trochanter and the femur.

It was formerly claimed by Bordage that while the autotomising middle and hind limbs could be regenerated by the immature insects, the fore-limbs possessed no such capacity. That his experiments on *Mantis religiosa* and other species failed to show regeneration was doubtless due to the age of the specimens taken, for Professor Przibram has since shown that the fore-limbs of immature *Mantis religiosa* and *Sphodromantis bioculata* normally regenerate if the operation be performed at a sufficiently early stage.

The theoretical interest of the points lies in the supposed correlation between the occurrence of autotomy and the capacity for regeneration, which Bordage's results were thought to indicate. It seemed desirable, therefore, to establish the regeneration of the fore-limb more firmly as a general rule for the Mantidae. It was therefore proposed to carry out experiments on two other forms, namely *Parathenodera angustifolia* and *Stagmomantis carolinensis*.

Further, it was thought that under some conditions there might be a tendency for a less differentiated form of limb to regenerate in the place of a removed fore-limb of the characteristic specialised type. Such a phenomenon would be of a piece with what occurs naturally in isolated cases in various Arthropods, and with what has been experimentally produced in a few forms by removing the ganglion supplying the appendage. It was therefore proposed to precede a series of fore-limb amputations in *Sphodromantis* by an operation on the thoracic ganglion on the same side.

Material and Methods.

The three species of Mantidae which were used in this work were *Sphodromantis bioculata*, *Parathenodera angustifolia*, and *Stagmomantis carolinensis*.

The *Sphodromantis* is the Sudanese form on which Professor Przibram has already done so much work, and the animals used were the offspring of some of those on which his experiments had been conducted.

The *Parathenodera* and the *Stagmomantis* were hatched out in the Institute from cocoons imported from Japan and North America respectively.

Throughout the experiments the animals were kept in muslin cages in a room the temperature of which was automatically maintained at 25 ° C. They were watered daily, and fed with flies as regularly as was possible.

The number of moults differs in the various forms but is not constant for any. The first moult always occurs immediately after hatching out: in *Parathenodera* maturity is reached after 7 or 8 (usually 8) moults, in *Stagmomantis* after 7 or 8, and in *Sphodromantis* after 8 (seldom), 9, 10 or more.

The amputations in all the experiments were made by clean cuts with scissors between the second and third joints (trochanter and femur). This corresponds to the place where autotomy occurs in the other limbs, and it has been shown that operation at that point in *Sphodromantis* is a favourable, although far from essential, condition for regeneration (Przibram). As a matter of routine the right foreleg was always chosen.

The work was unfortunately cut short before a satisfactory operation on the thoracic ganglion had been devised. The shortcomings of the operation used in the experiments to be described are indicated later.

Normal regeneration in Parathenodera.

For the first experiment a dozen individuals that had already passed the fifth moult were taken. With these it was possible to demonstrate the occurrence of regeneration, but not to obtain a regenerated limb of any great size, as this species becomes an imago at the seventh or eighth moult. It was intended to operate on a second series at an earlier stage, but the cocoons for these, possibly owing to their long refrigeration, never hatched. A few additional cases of regeneration were found as the result of accidental injuries among animals not under my care.

Details of Experiment I.

Material: *Parathenodera angustifolia* hatched on 23rd March, 1914, and moulted for the fifth time before 5th May.

Operation: Twelve individuals isolated on 5th May and their right forelegs amputated with scissors between the trochanter and the femur.

Progress: Eleven underwent their sixth moult on 6th and 7th May, the remaining one not until 13th May: one died shortly afterwards.

Ten moulted for the seventh time from 18th to 24th May (mean date, 21st May), one dying. The remaining specimen, doubtless the same that was late at the previous moult, did not undergo its seventh moult until 2nd June but then became a female imago (with an undeveloped right fore-wing). The remaining nine moulted for the eighth time from 5th to 21st June (mean date, 12th June), yielding two male and seven female imagines, one of each, however, dying in the moult.

Regeneration: Out of the original twelve two died as immature insects and one became an imago at the seventh moult. Of the nine that survived till an eighth and final moult two had recently injured the right coxa and thus lost the part of interest. Of the remaining seven only one failed to show some sign of regeneration, two showed slight regeneration, and four (two of each sex) showed good regeneration, the lost parts being replaced in miniature.

A very close comparison between the regenerated structures and the corresponding parts of the normal limb cannot fairly be made on account of the fact that the new growth was only of one or two moults standing and not comparable in size with the other limb, the right femur being not much more than a quarter of the length of the left.

But the parts of the regenerated limb seemed to be more cylindrical, the longitudinal ridges were much less marked, the spines were fewer in number, and the tarsus (where uninjured) had three or four joints instead of five. Of the four which showed good regeneration there was a male with four tarsal joints and a female with three, while the others had the tarsus injured.

Additional cases.

As already stated, the cocoons for a second experiment failed to hatch. It had been intended to operate on this series at an earlier stage and so to get regenerated structures more comparable in size with the normal parts. This purpose, however, was to some extent served by the occurrence of regeneration following accidental injuries among animals not specially isolated. The exact stage and the exact point at which the injury took place cannot, of course, be ascertained. Of four such cases one was a female and the others were males. The femur on the injured side was only ten per cent shorter than on the other, the discrepancy between regenerated and normal limb thus being scarcely obvious.

The only one which had the regenerated tarsus uninjured showed four joints in that part.

Normal Regeneration in Stagmomantis.

As in the case of *Parathenodera*, a series of operations was performed with animals which had already passed the fifth moult, and cocoons were taken from cold storage with a view to a second series at an earlier stage. In this case a few animals were hatched out and a second experiment was thus successfully carried through, although with very small numbers.

Details of Experiment II.

Material: *Stagmomantis carolinensis* hatched 13th and 14th April, 1914, and moulted for the fifth time on and after 22nd May.

Operation: Ten specimens isolated on 5th June and their right forelegs amputated with scissors between the trochanter and the femur.

Progress: Sixth moult from 6th to 13th June (mean date 9th June): four died in this moult or before the next.

Four specimens underwent a seventh but not final moult from 17th to 19th June, one dying soon after. For the other two the seventh moult was final and occurred later, a male imago emerging on 29th June and a female on 2nd July.

The remaining three moulted for the eighth time on 3rd and 4th July, all becoming female imagines.

Regeneration: Of the original ten only five survived to maturity. The only male, becoming an imago at the seventh moult, showed no regeneration. The four females, including the one for which the seventh moult was also final, all showed regeneration in some degree.

In two the regenerated structures were a perfect miniature of the normal parts except that in one case there were only four tarsal joints (tarsus injured in the other). In the other two the regeneration was small and very little differentiated: the existence of three segments (femur, tibia, tarsus) could, however, be made out in both, although the entire regenerated part in one case was only 1.5 mm. in length.

Details of Experiment III.

Material: *Stagmomantis carolinensis* hatched on 20th June, 1914, moulted for the second time about 1st July, and for the third from 6th to 9th July. (This cocoon was remarkable for liberating young insects on different dates, namely 12th, 17th and 20th June. The young insects which were hatched out on the two earlier dates did not long survive: the total for all three dates was under a score. The cocoon had been taken from cold storage on 3rd June and brought into a constant 25° C. next day).

Operation: Nine specimens isolated on 9th July and their right fore-legs amputated with scissors between the trochanter and the femur. Only four survived.

Progress: Fourth moult from 14th to 18th July.

Fifth moult from 23rd to 31st July.

One individual underwent a sixth moult on 4th August, but I had then to bring my work to an abrupt conclusion.

Regeneration: No regeneration was observed after the fourth moult, the one following the operation, nor in the first individual to undergo the fifth moult. The other three showed good regeneration after the fifth moult and the remaining one after the sixth. The regeneration displayed a well-developed replica of the lost parts in each case, but the specimens were among those that perished before a full description had been noted.

Regeneration after Operation on the Thoracic Ganglion in Sphodromantis.

In the experiments about to be described the usual amputation was preceded by an operation in which an attempt was made to destroy the thoracic ganglion on the same side by puncture with a hot needle. The only available criterion of success was the resulting paralysis of the right fore-leg: this limb was amputated in the usual way some hours later if the animals seemed otherwise uninjured. In all cases which ultimately gave any results the power of movement had returned, showing that enough ganglionic material had persisted to enable the motor nerve of the limb to be regenerated. The case was thus little different from the experiment designed as a control, in which the nerve was sectioned by cutting into the ventral border of the coxa (Experiment V).

Only one animal survived a series of attempts to dissect out the right thoracic ganglion under an anaesthetic, and further work at this was unfortunately impossible.

Details of Experiment IV.

Material: *Sphodromantis bioculata* moulted for the fourth time from 4th May, 1914, onwards.

Operation: Sixty specimens isolated and divided into three sets: —

1. Experiment: 23 specimens punctured with a hot needle in the region of the right thoracic ganglion, 20th May.

In 15 successful cases the paralysed right fore-leg was amputated with scissors between the trochanter and the femur on 21st May. Remaining seven individuals rejected.

2. Control A: 15 specimens without puncture operation had the right fore-leg amputated with scissors between the trochanter and the femur on 21st May.

3. Control B: 22 specimens kept as a normal control.

		Experiment	Control A	Control B
4th Moult	First	12. V. (23) Puncture operation 20. V. Amputation 21. V.	12. V. (15) Amputation 21. V. (15)	12. V. (22)
5th Moult	First	23. V.	22. V.	22. V.
	Mean	25. V. (9)	24. V. (14)	24. V. (22)
	Last	28. V.	25. V.	27. V.
6th Moult	First	6. VI.	1. VI.	1. VI.
	Mean	9. VI. (6)	6. VI. (12)	3. VI. (18)
	Last	14. VI.	11. VI.	7. VI.
7th Moult	First	20. VI.	16. VI.	11. VI.
	Mean	25. VI. (6)	23. VI. (12)	16. VI. (17)
	Last	28. VI.	29. VI.	23. VI.
8th Moult	First	3. VII.	30. VI.	24. VI.
	Mean	14. VII. (5)	9. VII. (11)	28. VI. (13)
	Last	22. VII. Two male imagines	23. VII. One male imago	2. VII. No imagines.
9th Moult (Incomplete on cessation of work)	First	29. VII.	27. VII. One male and one female imago	8. VII.
	Mean			17. VII. (7)
	Last			27. VII. One male and four female imagines

Progress: The progress of the experiment and the two controls may be expressed in tabular form. The first, last and mean dates for each moult are given, and the number of individuals surviving to reach each moult is stated in parenthesis after the mean date.

The retardation of moulting observable, especially in the experiment but also in the first control, may well be set down to the greater difficulty in catching flies in the case of animals which have been operated upon. The relatively large number of deaths recorded in the normal control is due partly to more active cannibalism and partly to the removal of specimens for other purposes.

Regeneration.

Experiment: On 23rd July, when five specimens survived after the eighth moult, it was noted that three showed regenerated limbs possessing some power of movement. One showed a less definite regeneration and the remaining individual had lost a further portion of the limb subsequent to the operation. Nothing abnormal was noted in the regenerated parts, but the material did not survive to undergo a minute examination. In any event, little special interest attaches to the case owing to the inadequacy of the puncture operation as shown by the return of movement to the limb.

Control A: On 23rd July, when ten specimens survived after the eighth moult, all showed normal regeneration. The normal regeneration for this species has already been described by Przibram.

Control B: (Normal animals not operated upon).

Details of Experiment V.

Material: *Sphodromantis bioculata* hatched on 28th April, 1914, and moulted for the fourth time from 27th May onwards.

Operation: Thirty-three specimens isolated on 3rd June and divided into two sets: —

1. Experiment: 22 specimens punctured with a hot needle in the region of the right thoracic ganglion, 3rd June. In 10 successful cases the paralysed right fore-leg was amputated with scissors between the trochanter and the femur on 4th June. The remaining twelve examples were rejected.

2. Control: 11 specimens had the right coxa cut with scissors to section the nerve, 4th June. In 9 successful cases the paralysed right fore-leg was amputated with scissors between the trochanter and the femur on 5th June. The remaining two individuals were rejected.

Progress: This may be tabulated as before.

	Experiment	Control
4th Moult	First 27. V. (22) Puncture operation 3. VI. Amputation 4. VI. (10)	27. V. (11) Nerve section 4. VI. Amputation 5. VI. (9)
5th Moult	First 7. VI. Mean 12. VI. (7) Last 17. VI.	7. VI. 8. VI. (9) 13. VI.

		Experiment		Control	
6th Moult	First	23. VI.		18. VI.	
	Mean	25. VI.	(4)	22. VI.	(9)
	Last	29. VI.		27. VI.	
7th Moult	First	3. VII.		27. VI.	
	Mean	4. VII.	(3)	2. VII.	(9)
	Last	4. VII.		6. VII.	
8th Moult (Incomplete on cessation of work)	First	17. VII.		8. VII.	
	Mean	18. VII.	(3)	17. VII.	(9)
	Last	18. VII.		29. VII.	
	No imagines			One male imago	

Regeneration.

Experiment: On 23rd July, when three specimens survived after the eighth moult, two showed regeneration of the lost parts and return of movement in the limb; the remaining individual had lost a further portion of the limb subsequent to the operation. The same remarks apply as in the previous experiment.

Control: On 29th July, when all nine specimens survived after the eighth moult, seven showed regeneration of the lost parts in various degrees and with movement of the limb. Of the remaining two one had lost a further portion of the limb subsequent to the operation. After a simple nerve section a regeneration of the motor nerve and a normal regeneration of the lost parts was to be expected; nothing contrary to this was brought to light in this control.

Note on Genital Stimulation.

During the attempts to contrive a satisfactory operation for the removal of the thoracic ganglion for the foregoing experiment a few adult male *Sphodromantis bioculata* were made the subjects of trial operations with and without anaesthetics. It was noted that there was a tendency for animals to assume the twisted attitude of copulation, and even to extrude spermatophores, while recovering from ganglion operations performed under an anaesthetic. The same effect was noted on anaesthetising an animal previously operated upon, but neither factor alone produced this result in any of these few cases.

Anaesthesia was produced by keeping the animal in a closed tube for some minutes along with a wad of cotton wool soaked with sulphuric ether. The animals were very apt to succumb altogether. All the animals were normal males except that No. 3 had been deprived of its antennae some time previously. The copulation of the Mantidae has been fully described by Przibram.

The effects of this kind which were noted may be summarised as follows:

No. 1:	a) 15. V. 1914. Unsuccessful puncture of the left thoracic ganglion with a cold needle	Negative
	b) 25. V. 1914. Nerve of right fore-leg cut away and nerve-cord severed on right side posterior to thoracic ganglion: Right ganglion probably injured	Negative
	c) 6 hours later: Cord severed on left also	Negative
	d) 27. V. 1914. Etherised	Nineteen hours later: in attitude of copulation and extruding a spermatophore
	e) 29. V. 1914. Again etherised	Fatal
No. 2:	a) 25. V. 1914. Etherised for 5 mins.	Negative
	b) Six hours later: Again etherised for 8 minutes and entire thoracic ganglia removed	Two days later: showed tendency to assume attitude of copulation
	c) 28. V. 1914. Again etherised	Same tendency during two following days. Ultimately fatal
No. 3:	a) 16. V. 1914. Unsuccessful puncture of left thoracic ganglion with unheated needle	
	b) Immediately afterwards: Etherised for 20 minutes and left thoracic ganglion removed, with injury also to right ganglion	Four hours later in attitude of copulation and spermatophore extruded: animal otherwise helpless
	c) 18. V. 1914. Again etherised for 10 minutes	Four hours afterwards, and during next day, in attitude of copulation with movement of genital armature; otherwise motionless. Ultimately died, after returning to normal attitude
No. 4:	a) 18. V. 1914. Etherised for 10 mins.	Negative during next two days: then fatal.

The evidence on this point affords very slight foundation on which to base theoretical considerations, but it may be suggested that the anaesthetic acted on a reflex mechanism already rendered excitable by the removal, through damage to the nerve-cord, of the control of the anterior portion of the nervous system. This explanation would be in accord with what is known on the subject in the case of higher animals, persistence of the genital reflex after transection of the spinal cord, and the occurrence of a phase of excitation during ether anaesthesia, being familiar phenomena.

Summary.

The Mantidae are characterised by the strong predatory fore-limbs which are not subject to autotomy. The middle and hind pairs of limbs both autotomise and regenerate.

Bordage found that amputation of the fore-limb was not followed by regeneration, and he concluded that the capacity for autotomy and the power of regeneration were correlated. He worked mainly with *Mantis religiosa,* probably taken at too late a stage.

Przibram found that amputation of the fore-limb at a sufficiently early stage was followed by regeneration both in *Mantis religiosa* and in the Sudanese *Sphodromantis bioculata.*

It was thought desirable to extend Przibram's experiments to other forms, those chosen being the Japanese *Parathenodera angustifolia* and the North American *Stagmomantis carolinensis.* Of twelve specimens of *Parathenodera* which had the right fore-leg amputated between the femur and the trochanter after the fifth moult, nine survived to become imagines at the eighth moult. Of these, two were injured and only one of the remainder failed to show some degree of regeneration. In several cases the regenerated structure was an almost perfect miniature replica of the lost parts, differing chiefly in the reduced number of tarsal joints (Experiment I). In the case of four not specially isolated individuals that had accidentally lost portions of fore-limb at an earlier stage, the regenerated parts were scarcely smaller than the corresponding parts of the opposite limb (Additional cases).

Of ten specimens of *Stagmomantis* similarly operated upon after the fifth moult, five survived to maturity at the seventh or eighth moults. Of these the only male showed no regeneration, but the four females showed regeneration in varying degrees. In two cases the regenerated structure was an almost perfect miniature reproduction of the normal parts: in one case a reduction in the number of tarsals was again noted, in the other the tarsus was injured (Experiment II).

Of four specimens of *Stagmomantis* which survived a similar operation after the third moult, three showed a well-developed regeneration of the lost parts after the fifth moult, the remaining one showing the same after the sixth moult. The experiment could not be carried further (Experiment III).

The substitution of a less specialised for a more complex appendage occurs in nature as an occasional abnormality in many Arthropods. The same thing has been experimentally produced in a few cases by extirpation of the ganglion supplying the part. It was therefore decided to observe the effect, on the regeneration of the fore-limb of *Sphodromantis bioculata*, of an operation on the thoracic ganglion on the same side previous to the amputation of the limb below the trochanter.

Two series of *Sphodromantis* were subjected to the usual amputation operation following a puncture of the thoracic ganglion on the same side with a hot needle. Although normal regeneration followed in almost all surviving cases the interest of these was destroyed by the return of movement to the limb, indicating the failure of the ganglion operation. For one of the experiments a control series of simple amputation cases and a control series of normal animals were kept. In the other a control series was kept in which the amputation had been preceded by a section of the nerve in the coxa: return of movement and normal regeneration followed. The normal regeneration in this form had already been described by Przibram and nothing out of accord with his description was noted (Experiments IV and V). Only one animal survived a dissection of the right thoracic ganglion under an anaesthetic and the work could not, at the time, be carried further. Trial operations on the thoracic ganglia of adult male *Sphodromantis bioculata* anaesthetised with sulphuric ether, caused the subjects to assume the attitude of copulation and even to extrude spermatophores (Note on Genital Stimulation).

(Deutsche Zusammenfassung im Akad. Anzeiger Wien, Nr. 7, 8, 1921.)

References.

Bordage, E. (1899): »Régénération des Membres chez les Mantides«, Compt. Rend. Acad. Paris, 128, 1593. — Idem (1905): »Recherches anatomiques et biologiques sur l'Autotomie et la Régénération chez divers Arthropodes«, Thèses Fac. Sciences Paris, Sér. A, 494, No. 1207. — Przibram, H. (1906): »Aufzucht, Farbwechsel und Regeneration einer ägyptischen Gottesanbeterin«, Arch. f. Entw.-Mech., 22, 149. — Idem (1907): »Aufzucht, Farbwechsel und Regeneration unsrer europäischen Gottesanbeterin«, Arch. f. Entw.-Mech., 23, 599. — Idem (1907): »Paarungsversuche an Gottesanbeterinnen«, Verh. Morph.-physiol. Ges. Wien, Zentralblatt f. Physiol., 221. Nr. 8. — Idem (1909): »Aufzucht, Farbwechsel und Regeneration der Gottesanbeterinnen, III«, Arch. f. Entw.-Mech., 28, 561.